Discussion 6: Part II — Review

Work with your group to complete this review before attempting Problems 4–6. Fill in the blanks and discuss the reasoning. The examples below are separate from the worksheet questions.

A. Sampling distribution of a sample proportion

1. A survey records whether each person supports a proposal. Let p be the proportion of *all* people who support it, and let x out of n surveyed people support it. The observed sample proportion is $\hat{p} =$ _____. The population proportion is a _____, while the sample proportion varies from sample to sample.

2. Let X be the number of supporters among n independent people, each with the same probability p of supporting the proposal. Then $X \sim$ _____ and $\hat{P} = X/n$. The mean of $\hat{P}$ is $\mu_{\hat{P}} =$ _____ and its standard deviation is $\sigma_{\hat{P}} =$ _____.

3. Before using a normal approximation for $\hat{P}$, check _____ ≥ 10 and _____ ≥ 10. These conditions ensure sufficiently many expected _____ and _____.

B. Planning the sample size for a confidence interval

4. A confidence interval estimates an unknown population mean. Its *margin of error* (ME) tells us how far the endpoints are from the sample mean. For a z-confidence interval with known population standard deviation σ, write the formula: $ME =$ _____.

5. Sometimes we want to decide *how many observations to collect* before conducting a study. We choose a sample size n large enough to achieve a desired margin of error. Using the formula above, write the formula for the required sample size: $n =$ _____.

Tip: Always round the required sample size up to a whole number so the margin of error is no larger than requested.

C. Testing a claim about a population mean

6. A delivery service claims its population mean delivery time is *greater than* 25 minutes. Let μ be the population mean delivery time. Write the alternative hypothesis H_A : _____ and the null hypothesis H_0 : _____.

7. The direction of H_A tells us which values provide evidence for the claim. Complete the descriptions: $H_A : \mu < \mu_0$ means we look for unusually _____ sample means; $H_A : \mu > \mu_0$ means unusually _____ sample means; $H_A : \mu \neq \mu_0$ means unusually _____ *or* _____ sample means.

8. When the population standard deviation is known, the test statistic for a population mean is $z =$ _____. What quantity appears in its denominator? _____

9. A p-value measures how unusual the observed result (or a more extreme result) would be if the null hypothesis were true. If p-value $< \alpha$, we _____ H_0; otherwise, we _____ H_0.

10. After making a decision, explain what it means *in the context of the original claim*. If we reject H_0, we have _____ evidence supporting H_A. If we fail to reject H_0, we do _____ have sufficient evidence supporting H_A.

Step 1: write down $\begin{cases} H_A. \rightarrow \text{claim which we want to test} \\ H_0. \end{cases}$

Step 2: check left / right / two.

Step 3: calculate $Z = \dfrac{\bar{X} - u}{\frac{\sigma}{\sqrt{n}}}$.

Step 4: calculate p-value.

Step 5: check $p < \alpha$. $\rightarrow$ 0.1 0.05 0.02.

Step 6: reject / fail to reject H_0

Step 7: _____. say something.

Discussion 6: Part II — Worksheet (Problems 4–5)

4. Thirty-five percent of single-family homes have a garage. A realtor selects a random sample of 250 single-family homes, and finds that 77 have a garage. $p = 0.35$, $n = 250$, $X = 77$

a) Calculate $\hat{p}$.

$\hat{p} = \frac{X}{n} = \frac{77}{250} = 0.308$.

b) Define the random variable of interest, $\hat{P}$.

$\hat{P}$ is the sample proportion of 250 selected single-family homes.

c) State the approximate distribution of $\hat{P}$. Check the relevant criteria.

X: # of homes w/ a garage.

Check Binomial criteria: ① fixed $n = 25$ ② independent observations because they are random examples ③ fixed $p = 0.35$ ④ two outcomes.

so $X \sim Bin(250, 0.35)$. check $\begin{cases} np = 87.5 > 10 \\ n(1-p) = 162.5 > 10 \end{cases}$

so $\hat{p} \overset{approx}{\sim} N\left(p, \sqrt{\frac{p(1-p)}{n}}\right)$ which is $N(0.35, 0.0302)$.

5. The variance of the lengths of betta fish is assumed to be 1.96 cm^2. How large of a sample would you need to estimate the population mean length of a betta fish with 90% confidence and a margin of error of 0.08 cm? Assume the length of a betta fish is known to be normally distributed.

$\sigma^2 = 1.96$ so $\sigma = 1.4$. $\alpha = 0.1$.

$ME = 0.08 = z_{\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{n}}$

so $n = \left(\frac{z_{\frac{\alpha}{2}} \cdot \sigma}{ME}\right)^2 = 828.68$

round up: $n \approx 829$.

So we should choose at least 829 samples.

Discussion 6: Part II — Worksheet (Problem 6)

6. The advertisement for a new board game claims that the average length of game play is less than an hour. In a random sample of 90 game plays, the mean length is calculated to be 56.9 minutes. If we assume that the population standard deviation (of the length of game play) is 14.5 minutes, test the advertisement's claim at a 5% significance level.

$H_A: \mu < 60$ $n = 90$ $\bar{x} = 56.9$ $\sigma = 14.5$ $\alpha = 0.05$

a) Define the parameter and random variable of interest.

μ: population mean game play time

$\bar{X}$: sample mean time of a random sample of 90 game plays.

b) State the null and alternative hypotheses, and identify the claim.

$H_0: \mu \geq 60$

$H_A: \mu < 60$ → this is the claim!

c) Determine the distribution of the test statistic. (Check the relevant criteria.)

Population distribution is unknown. Since $n = 90 > 30$, the distribution of sample mean is approximately normal.

Also, population sd $\sigma = 14.5$ is known. So we use z-test.

d) Calculate the test statistic.

under H_0: $\mu_0 = 60$, the test statistic is approximately normal.

$$z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}} \approx -2.03.$$

e) Find the p-value.

$H_A: \mu < 60$. so we use left tail. p-value $= P(Z < -2.03) \approx 0.0212$.

f) State your decision.

$p = 0.0212 < 0.05$ So we reject H_0.

g) State your conclusion.

At the 5% significance level, there is sufficient evidence to support the advertisement's claim that population mean is less than 60 minutes.